
Diffusion-Guided Robustness: Leveraging Generative Modeling for Defense Against Adversarial Attacks in Financial Fraud Detection Systems

Atharva Laud
University of British Columbia
alaud@student.ubc.ca

Abstract

1 The critical reliance of modern financial systems on Machine Learning (ML) for
2 real-time fraud detection has exposed them to sophisticated Adversarial AI (AAI)
3 attacks. To counter these threats, particularly evasion attacks, this paper explores
4 the application of Denoising Diffusion Probabilistic Models (DMs) as a superior
5 generative primitive for improving adversarial training robustness. The central
6 hypothesis is that DMs, with necessary architectural adaptations for tabular data
7 and extreme class imbalance, provide an efficient and effective framework through
8 two primary defense vectors: post-hoc adversarial purification and robust synthetic
9 data augmentation. Modifications such as unified diffusion frameworks (TabDiff,
10 TabDDPM) and regularization techniques (Contrastive Learning, CBDM) will be
11 employed, to handle mixed-type tabular data and mitigate mode collapse. Further-
12 more, DMs can serve as essential attack simulators (e.g., Diff-PGD) to generate
13 distributionally realistic, stealthy threats for continuous defense hardening. The
14 analysis confirms DMs' significant potential for building robust, high-throughput,
15 and low-latency defense systems in the FinTech industry.

16 1 Introduction: The Imperative for Adversarial Robustness in FinTech

17 1.1 The Critical Vulnerability of ML Models in Financial Systems

18 Modern financial institutions depend heavily on sophisticated Machine Learning (ML) models for
19 real-time fraud detection [18]. While fast and efficient, their complexity introduces critical security
20 vulnerabilities exploited through Adversarial AI (AAI) [11]. AAI attacks primarily include data
21 poisoning (compromising the training dataset) and evasion attacks (crafting malicious inputs to
22 circumvent real-time monitoring) [10]. Attackers often possess significant knowledge, allowing for
23 highly targeted evasion strategies against models designed to detect financial crimes such as credit
24 card fraud.

25 1.2 The Case for Diffusion Models as a Robust Generative Primitive

26 Countering sophisticated adversarial threats requires advanced generative modeling techniques to
27 stress-test and harden ML defenses. While Generative Adversarial Networks (GANs) and Variational
28 Autoencoders (VAEs) have been explored, Denoising Diffusion Probabilistic Models (DMs) have
29 emerged as superior alternatives, overcoming limitations like training instability, mode collapse,
30 and poor representation of multimodal distributions [19]. The DM mechanism—a forward process
31 that adds Gaussian noise and a reverse process that learns to denoise it [1]—enables high-fidelity
32 and diverse modeling of complex data manifolds [15]. This capability is essential for accurately
33 generating rare fraud scenarios and for purifying corrupted inputs back to the clean financial dataset.

34 1.3 Hypothesis and Structure

35 The hypothesis is that Diffusion Models, through necessary architectural adaptations for financial
36 data’s specific constraints (tabular structure and extreme class imbalance), offer an efficient and
37 effective framework for enhancing adversarial robustness. This is achieved through post-hoc adver-
38 sarial purification and robust synthetic data augmentation. The following sections detail the threat
39 landscape, the required architecture, and the deployment of DMs as both a defense mechanism and
40 an attack simulator.

41 2 Background and Related Work: The Adversarial Threat Landscape and 42 Generative Modeling in Security

43 2.1 Defining the Adversarial Threat in Financial Operations

44 Adversarial attacks in finance aim to undermine the integrity of the fraud detection system via
45 poisoning or real-time evasion [18, 11, 10]. A fundamental quality common among most uncaught
46 fraud cases success is stealth. Traditional gradient-based techniques often produce noticeably
47 perturbed inputs that violate statistical norms (e.g., spending patterns), risking detection by simple
48 heuristic rules before reaching the ML classifier [5]. Therefore, any generative method used for
49 simulating or defending against attacks must strictly adhere to the physical constraints of the financial
50 transaction manifold. The DM’s core strength lies in its ability to accurately learn the true distribution
51 of clean data [1], allowing it to generate distributionally sound adversarial examples, such as those
52 produced by the Diffusion-Based Projected Gradient Descent (Diff-PGD) framework [16].

53 2.2 Generative Models for Anomaly Detection (AD) and Robustness

54 Generative models are foundational in anomaly detection. Empirical evaluations have shown that
55 diffusion-based anomaly detection methods perform competitively and can achieve faster infer-
56 ence times than standard DDPMs, positioning them as scalable alternatives [14]. DMs are the
57 preferred choice when the generation task demands high output fidelity and diversity [15], which
58 is non-negotiable for synthesizing complex, rare fraud events necessary for security augmentation.
59 Continuous advancements, such as fast non-Markovian diffusion models (FNDM) [14], continue to
60 improve their practical value in high-precision fraud detection.

61 3 Architectural Challenges: Tailoring Diffusion Models for Tabular Finance 62 Data

63 The successful deployment of DMs in financial fraud detection requires overcoming the challenges
64 of tabular data structure and extreme class imbalance.

65 3.1 Modeling Mixed-Type Tabular Data Complexity

66 Financial datasets are fundamentally mixed-type tabular data, comprising numerical, categorical,
67 and other feature types [9]. This heterogeneity is more challenging to model than homogeneous
68 data like images, and approaches combining separate generative models for different feature types
69 fail to capture complex, joint dependencies [9]. Consequently, solutions must employ unified
70 diffusion frameworks. Models like TabDiff [8] and TabDDPM [13] address this by normalizing
71 numerical columns and converting categorical features to one-hot vectors. A joint forward diffusion
72 process is then applied across all modalities, which is vital for maintaining plausible pair-wise
73 column correlations—a crucial measure of data quality and integrity that ensures the synthetic data is
74 statistically valid [8, 9].

75 3.2 Addressing Extreme Class Imbalance and Mode Collapse

76 Financial fraud data has an extreme long-tailed distribution, where non-fraud instances vastly outnumber
77 fraudulent ones [17]. When DMs are trained on such data, a significant degradation in diversity
78 and fidelity occurs in the rare tail classes, leading to mode collapse [17]. This results in synthetic

79 fraud examples that are redundant and limited, restricting the detection model’s ability to generalize
80 to novel evasion attacks. Specialized architectural adjustments are necessary to mitigate this bias [12].
81 Two primary solutions are leveraged:

- 82 • **Contrastive Learning (CL) Integration:** CL regularization forces the DM to better learn
83 the distinct characteristics of the rare fraud manifold by separating dissimilar instances in
84 the embedding space, thus enhancing minority class generation diversity [3, 4].
- 85 • **Distribution Adjustment (CBDMM):** Class-Balancing Diffusion Models (CBDMM) use a
86 distribution adjustment regularizer to minimize the distance between the estimated noise
87 of the original DDPM and a uniform distribution, yielding higher quality and diversity in
88 minority class generation [17].

89 4 Diffusion Models as a Defense Framework against Adversarial Fraud

90 DMs offer two powerful, complementary modalities for enhancing model robustness: purification
91 and augmentation.

92 4.1 Defense via Efficient Adversarial Purification

93 Adversarial purification utilizes the DM’s reverse generation process as a highly effective preprocessor.
94 The principle is to map an adversarial input, which is perturbed off the clean data manifold, back
95 toward the manifold learned from clean data, thereby denoising the adversarial perturbation before it
96 reaches the classification layer [6]. This strategy is computationally efficient, as it avoids the high
97 cost of continual adversarial retraining and does not require gradient computations during the defense
98 step [6]. This efficiency is vital for the low-latency requirements of real-time financial systems [18].
99 Key techniques include:

- 100 • **Adversarial Anti-Aliasing (AAA):** A preprocessing technique to reduce the magnitude
101 of feature-level perturbations, effective against both perturbation-based and stealthy, unre-
102 stricted adversarial attacks [6, 7].
- 103 • **Adversarial Super-Resolution (ASR):** Leverages the DM’s extensive prior knowledge of
104 clean datasets to benignly recover the high-quality structure of the transaction data from the
105 adversarially degraded version [6].

106 4.2 Defense via Synthetic Data Augmentation and Robust Training

107 DMs are transformative tools for generating high-quality synthetic data, which is essential for
108 hardening ML models [12]. By utilizing CL-enhanced DM architectures, institutions can strategically
109 augment the tail of the distribution with synthetic fraud samples that possess high diversity and
110 fidelity [3]. This targeted inoculation trains the core detection model to recognize a significantly
111 broader range of subtle, novel adversarial modifications, fundamentally expanding the model’s
112 generalization capabilities and sensitivity to emerging attack vectors [12].

113 5 Diffusion Models as an Attack Simulator: Stress Testing the Defense 114 Architecture

115 To rigorously validate any DM-based defense framework, the testing process must employ adversarial
116 threats that are both effective and highly realistic. DMs are essential attack simulators because they
117 can generate these sophisticated threats.

118 5.1 Generating Distributionally Realistic Adversarial Examples

119 The Diffusion-Based Projected Gradient Descent (Diff-PGD) framework generates realistic adversar-
120 ial samples by leveraging the DM’s learned data distribution as a gradient guide [16]. This process
121 ensures the crafted adversarial inputs remain exceptionally close to the true data manifold, optimizing
122 attack effectiveness while maintaining the stealthiness required for financial evasion [5]. The result-
123 ing samples exhibit superior transferability and "anti-purification power" compared to traditional
124 methods [5].

5.2 The Feedback Loop: DM-Attack informs DM-Defense

Deploying DMs for attack simulation establishes a continuous feedback loop. DM-guided adversarial examples identify blind spots within the purification network and the regularization schemes (CL/CBDM) used for synthetic augmentation. The insights gained allow for targeted iterative improvements to the generative and restorative components of the defense architecture, ensuring the continuous hardening of the overall fraud detection system against the most sophisticated threats.

6 Expected Results

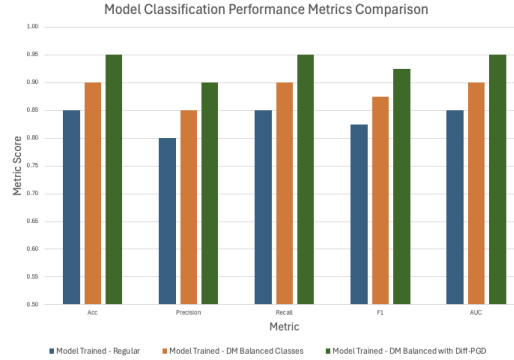


Figure 1: Expected classification model results from different training methods using Diffusion

A hidden metric is the model’s ability to handle completely new and "unseen" cases of fraud and report it. Regular fraud-detection models would most likely struggle with this, however the DM balanced and Diff-PGD model would excel in these cases as a result of greater training data variety and quantity.

7 Conclusion and Future Directions

Diffusion models represent a paradigm shift in generative modeling applied to adversarial defense, offering a compelling combination of high-fidelity data synthesis and efficient, non-retraining based purification mechanisms. This makes them exceptionally well-suited for the demanding, high-stakes, and computationally constrained environment of real-time financial fraud detection. The adoption of DM-guided attack simulation (Diff-PGD) ensures that defenses are validated against distributionally realistic, stealthy threats, closing the loop on continuous security hardening.

Future research must focus on optimizing the operational deployment of DM-based defense mechanisms:

- **Optimizing Real-Time Inference Speed:** The inference speed of standard DDPMs remains a limitation for low-latency systems [15]. Future investigations must prioritize the integration and empirical evaluation of faster diffusion variants, such as Denoising Transformer Encoders (DTE) or fast non-Markovian diffusion models (FNDM), to ensure high throughput without compromising purification fidelity [14].
- **Refining Tabular Modeling for Discrete Data:** The challenge of modeling high-cardinality categorical data remains a barrier [9]. Continued advancements in unified discrete diffusion models are required to move beyond simple encoding methods which risk losing semantic information during diffusion [2].
- **Benchmarking Purification Efficacy on Tabular Data:** Comprehensive empirical studies are urgently needed to quantify the protective effects of proposed purification methods (AAA and ASR) when applied to complex, non-image tabular feature spaces, confirming their efficiency against the unique characteristics of tabular evasion attacks [7].

References

- [1] AdvDiff Authors. Advdiff: Generating unrestricted adversarial examples using diffusion models. *arXiv preprint arXiv:2307.12499v4*, 2023. Accessed: 2025-10-09.
- [2] CADDM Authors. [2510.01329] continuously augmented discrete diffusion model for categorical generative modeling. *arXiv preprint arXiv:2510.01329*, 2025. Accessed: 2025-10-09.
- [3] CLIDM Authors. Can contrastive learning improve class-imbalanced diffusion model? *arXiv preprint arXiv:2507.09052v1*, 2025. Accessed: 2025-10-09.
- [4] CLIDM Abs Authors. [2507.09052] can contrastive learning improve class-imbalanced diffusion model? *arXiv preprint arXiv:2507.09052*, 2025. Accessed: 2025-10-09.
- [5] Diff-Based Authors. Diffusion-based adversarial sample generation for improved stealthiness and controllability. *arXiv preprint arXiv:2305.16494v3*, 2023. Accessed: 2025-10-09.
- [6] Purification Authors. [2501.13336] gradient-free adversarial purification with diffusion models. *arXiv preprint arXiv:2501.13336*, 2025. Accessed: 2025-10-09.
- [7] Purification HTML Authors. Gradient-free adversarial purification with diffusion models. *arXiv preprint arXiv:2501.13336v1*, 2025. Accessed: 2025-10-09.
- [8] TabDiff Authors. Tabdiff: a mixed-type diffusion model for tabular data generation. *arXiv preprint arXiv:2410.20626v3*, 2024. Accessed: 2025-10-09.
- [9] TabularDM Authors. Diffusion models for tabular data: Challenges, current progress, and future directions. *arXiv preprint arXiv:2502.17119v1*, 2025. Accessed: 2025-10-09.
- [10] Bruno Carminati, Elena Ferrari, and et al. Evasion attacks against banking fraud detection systems. In *Proceedings of the USENIX Symposium on Networked Systems Design and Implementation (NSDI)*. Accessed: 2025-10-09.
- [11] CrowdStrike. What is data poisoning? <https://www.crowdstrike.com/en-us/cybersecurity-101/cyberattacks/data-poisoning/>. Accessed: 2025-10-09.
- [12] FCA. Report: Using synthetic data in financial services. <https://www.fca.org.uk/publication/corporate/report-using-synthetic-data-in-financial-services.pdf>. Accessed: 2025-10-09.
- [13] Anton Kotelnikov and Dmitry Kotelnikov. Tabddpm: Modelling tabular data with diffusion models. In *Proceedings of Machine Learning Research*, 2023. Accessed: 2025-10-09.
- [14] Fangdong Liu. fdjingliu/DMAD: The collection of diffusion models for anomaly detection, a survey paper submitted to IJCAI 2025. <https://github.com/fdjingliu/DMAD>. Accessed: 2025-10-09.
- [15] Milvus. How does a diffusion model compare with gans and vaes? <https://milvus.io/ai-quick-reference/how-does-a-diffusion-model-compare-with-gans-and-vaes>. Accessed: 2025-10-09.
- [16] OpenReview. Diffusion-based adversarial sample generation for improved stealthiness and controllability. [https://openreview.net/forum?id=JTwxyLP6U9&referrer=%5Bthe%20profile%20of%20Yongxin%20Chen%5D\(%2Fprofile%3Fid%3D~Yongxin_Chen1\)](https://openreview.net/forum?id=JTwxyLP6U9&referrer=%5Bthe%20profile%20of%20Yongxin%20Chen%5D(%2Fprofile%3Fid%3D~Yongxin_Chen1)). Accessed: 2025-10-09.
- [17] Zheng Qin and et al. Class-balancing diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. Accessed: 2025-10-09.
- [18] ResearchGate. Adversarial attack detection using explainable ai and generative models in real-time financial fraud monitoring systems. https://www.researchgate.net/publication/393605167_Adversarial_Attack_Detection_Using_Explainable_AI_and_Generative_Models_in_Real-Time_Financial_Fraud_Monitoring_Systems. Accessed: 2025-10-09.
- [19] ResearchGate. Diffusion models for tabular data: Challenges, current progress, and future directions. https://www.researchgate.net/publication/389316416_Diffusion_Models_for_Tabular_Data_Challenges_Current_Progress_and_Future_Directions. Accessed: 2025-10-09.

A Appendix

This appendix provides a detailed overview of the external dataset utilized for the models presented in this paper: the IEEE-CIS Fraud Detection dataset, sourced from a Kaggle competition hosted by the IEEE Computational Intelligence Society (IEEE-CIS) and Vesta Corporation. The dataset comprises real-world e-commerce transactions and is designed to challenge machine learning models in predicting the probability of an online transaction being fraudulent.

A.1 Data Structure and Contents

The dataset is extensive and is delivered in four separate files for both the training and testing phases:

1. `train_transaction.csv` and `test_transaction.csv`: These files contain the core transaction information, including the target variable, transaction amount, product code, payment card details, and numerous masked features.
2. `train_identity.csv` and `test_identity.csv`: These files contain supplemental identity information—such as network connection details (IP, ISP), device type, and browser information—for a subset of the transactions.

The two main data tables can be merged using a common key: `TransactionID`. Not all transactions have corresponding identity information, which is a key characteristic of the data requiring specific preprocessing.

A.2 Key Features and Characteristics

The dataset contains a rich, anonymized set of features, many of which have been masked for privacy and security. These features can be broadly categorized as follows:

- **Target Variable** (`isFraud`): A binary indicator (0 or 1) representing whether the transaction is fraudulent.
- **Time Feature** (`TransactionDT`): This feature represents a `timedelta` from an arbitrary reference date, allowing for time-series analysis of transaction activity.
- **Masked Features** (e.g., `V1-V339`, `id_01-id_38`): These numerical and categorical features were engineered by Vesta to capture complex relationships and details regarding device characteristics, transaction velocity, and payment-related counters. These features often exhibit a high degree of missing values and high inter-correlation, necessitating careful feature engineering and imputation strategies.
- **Categorical Features**: Includes non-masked fields such as `ProductCD`, card details (`card1-card6`), and email domains (`P_emaildomain`, `R_emaildomain`).

The substantial volume of both missing data and high-dimensionality in the masked V-features poses a significant challenge, requiring robust data cleaning and dimensionality reduction techniques.